

# BACL Tech Spec Report

## Technical Specification: Bio-Acoustic Cryptographic Ledger (BACL)

### Project Overview

A decentralized encryption framework leveraging biological soundscapes from complex ecosystems as the entropy source for cryptographic key generation.

### System Architecture

#### 1. Data Acquisition Layer (The "Ear")

- **Hardware:** High-fidelity omnidirectional microphones/hydrophones (10Hz – 48kHz).
- **Input:** Live organic signals from protected biomes.
- **Filtering:** Band-pass filtering to isolate organic signals from anthropogenic noise.

#### 2. Entropy Extraction (The "Nexus")

- **FFT:** Conversion of audio to frequency-domain data.
- **Spectral Entropy:** Measuring unpredictability at millisecond intervals.
- **Hashing:** Processing through **SHA-3 (Keccak)** or **BLAKE3** for a 256-bit seed.

#### 3. Multi-Node Fusion (The "Constellation" Protocol)

- **Geographic Sharding:** Requires synchronized entropy from geographically distant nodes (e.g., forest and marine biomes).
- **XOR Fusion:** Final master key generated via bitwise XOR of hashes from multiple nodes ( $H_1 \oplus H_2$ ).
- **Time-Sync:** Precise millisecond synchronization via NTP.

#### 4. Dynamic Redundancy (The "Storm-Guard" Fail-Safe)

- **Watchdog:** Monitors Shannon Entropy ( $H < 7.9$  bits/byte).
- **Secondary Source:** Automatically pivots to atmospheric noise or hardware TRNG if biological entropy drops.

### Technical Specifications

- **Encryption Standard:** AES-256-GCM (Authenticated Encryption).
- **Key Space:**  $2^{256}$  (~ $1.1 \times 10^{77}$  combinations).
- **Languages:** **Rust** (Edge Nodes), **Go** (Networking), **Python** (Crypto/API).

### Security Logic

- **Quantum Resistance:** AES-256 is highly resilient to near-future quantum attacks.
- **Anti-Spoofing:** Secondary salts (temp/barometric data) and ML-verification prevent audio injection/replay attacks.